First-Come, First-Served (FCFS) CPU Scheduling

Core Scheduling Policy

- Non-preemptive execution backed by a First-In, First-Out (FIFO) ready queue.
- Running process retains the CPU until its CPU burst completes or blocks on I/O.
- Zero starvation, but average waiting time depends heavily on arrival order.

The Convoy Effect

- One long CPU-bound process arrives first and stalls many short processes behind it.
- Reduces overall CPU and I/O device utilization compared to shorter-first policies.

Turnaround, Waiting Time Formulas & Worked Example

Scheduling Equations

- Turnaround Time: TAT = Completion Time (CT) - Arrival Time (AT).
- Waiting Time: WT = Turnaround Time (TAT) - Burst Time (BT).

Worked Gantt Chart: [0 | P1 | 4 | P2 | 7 | P3 | 9]

- P1 (AT=0, BT=4): CT=4 ms, TAT=4 ms, WT=0 ms.
- P2 (AT=1, BT=3 inferred from [4..7]): CT=7 ms, TAT=6 ms, WT=3 ms.
- P3 (AT=2, BT=2): CT=9 ms, TAT=7 ms, WT=5 ms -> Avg WT = 2.67 ms, Avg TAT = 5.67 ms.